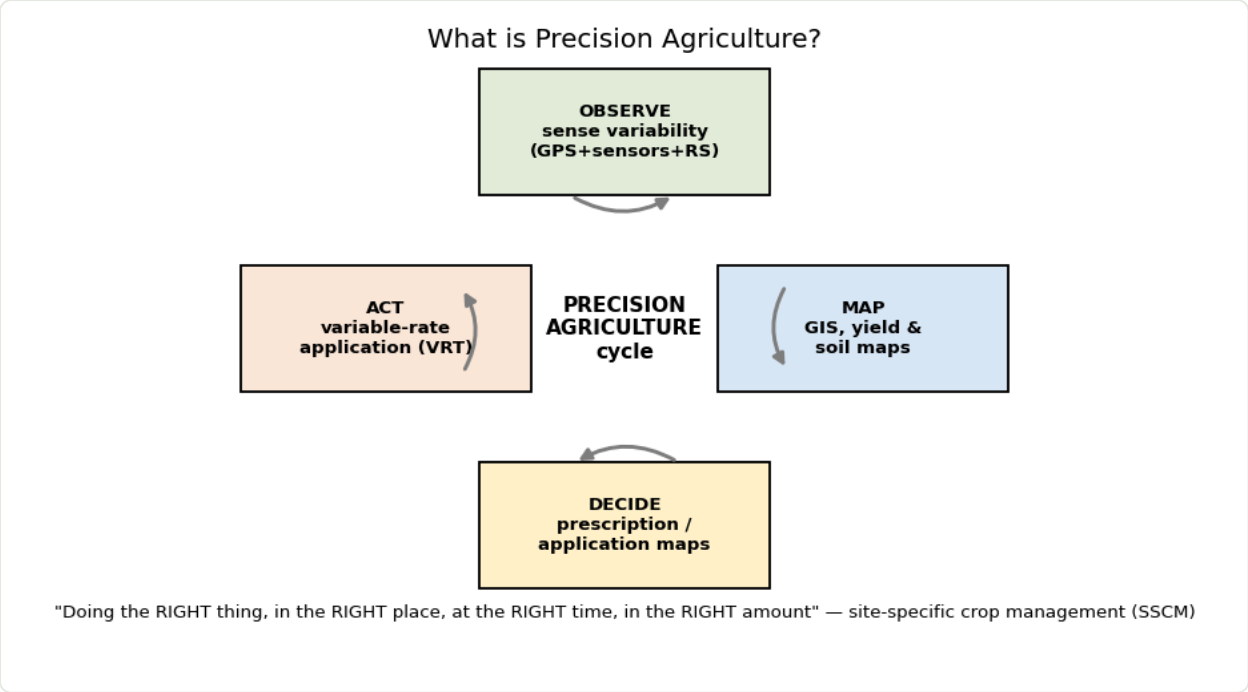
UNIT 10 · TOPIC 1

Importance of Precision Agriculture

Syllabus topic: Importance of precision agriculture; mapping in farming for decision-making

What precision agriculture is

PRECISION AGRICULTURE (PA) — also called site-specific crop management (SSCM) — manages a field according to its WITHIN-FIELD VARIABILITY instead of treating it uniformly: the right input, in the right place, at the right time, in the right amount.



The precision-agriculture cycle

The PA cycle

1. **OBSERVE** — sense variability with GPS, sensors and remote sensing (Topics 3, 6).
2. **MAP** — GIS, yield and soil maps (Topics 4, 5, 8).
3. **DECIDE** — prescription/application maps (Topic 8).
4. **ACT** — variable-rate application (Topic 9); then repeat.

Why it matters

- Raises INPUT-USE EFFICIENCY — seed, fertiliser, water and chemicals go only where needed → lower cost, less waste.
- Increases YIELD and PROFIT.
- Cuts ENVIRONMENTAL impact — less nutrient/chemical run-off (Topic 11).

Enabling technologies: GPS/GNSS, GIS, remote sensing, proximal sensors, yield monitors, variable-rate technology (VRT) and controllers, and data analytics (ML/IoT — Unit 2, Topic 10).

Practice MCQs

Q1. Precision agriculture means managing a field according to its:

(a) Uniform average
(b) Within-field variability (site-specific)
(c) Colour only
(d) Owner

Q2. The PA management cycle is:

(a) Act → observe
(b) Observe → map → decide → act
(c) Harvest → sow
(d) Buy → sell

Q3. A key benefit of precision agriculture is:

(a) Higher uniform input use
(b) Better input-use efficiency with lower cost and environmental impact
(c) More labour
(d) Larger fields

Q4. Which is NOT an enabling technology of PA?

(a) GPS/GNSS
(b) GIS & remote sensing
(c) Variable-rate technology
(d) Manual broadcasting by hand

Answers

Q1 — (b) Within-field variability (site-specific)
Q2 — (b) Observe → map → decide → act
Q3 — (b) Better input-use efficiency with lower cost and environmental impact
Q4 — (d) Manual broadcasting by hand

Sources: Srinivasan — Handbook of Precision Agriculture; NIIT/ICAR precision-agriculture modules; NovAtel — Introduction to GNSS; ISRO/PIB NavIC data (as given in the source notes, July 2026; operational status updated October 2026); NPTEL precision-agriculture lectures.